

Wilson, Chow Sin Long

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Senior Product Frontend Engineer · AWS Certified Solutions Architect – Associate

SUMMARY

Senior product frontend engineer with 5 years of experience shipping 15+ React and React Native products from Figma and design systems through release, including App Store and Google Play. Cut load times by 80% and search latency by 90%. Now building the chat, mentions, and resume-mid-run UI for a multi-channel **agentic** platform. AWS Certified Solutions Architect – Associate.

EXPERIENCE

Frontend Developer, Vioo Company Limited

May 2022 – Present

Primary frontend engineer for concurrent client products, scoping UX with designers and stakeholders, turning Figma into tokenized, composable UI, and owning each product flow through release.

- Shipped and maintained **15+** React / React Native apps with high-fidelity Figma design, shared design systems, and Apple App Store / Google Play releases.
- Built **design systems** on **Radix** / **Tailwind** / **shadcn/ui**: themed with CSS **tokens** (color, type, space); Built composable components and shared hooks. Documented with **Storybook**, **Fumadocs**, MDX.
- Migrated WordPress across client sites to **headless CMS** platforms (Strapi / Sanity) and **unified search** (Meilisearch / Algolia) in a **Turborepo** on **Next.js** / **SST**; cut load times by **80%** and search latency by **90%**, and achieved PageSpeed Insights scores above **95**.
- Established **Vitest** and **Playwright** / **Cypress** as quality gates for shared UI and the product flows using it.
- Instrumented client web and mobile apps and their APIs with **Sentry**, making production crashes, device issues, and API errors visible after release.
- Led adoption of **tRPC** and Scalar-generated API documentation, cutting API integration time by **50%** and eliminating schema-mismatch errors that blocked product work.
- Built GitHub Actions + **SST** CI/CD with linting, type-checking, tests, and builds; cut deployment time by **50%+**.
- Mentored summer interns and junior developers through pair programming, code reviews, and guidance as they shipped production UI.

SELECTED PROJECTS

Multi-Channel Agentic Platform

Apr 2026 – Present

Chat and workspace UI for a durable agent that powers internal tools and workflows.

- Built agent runs durable streams so clients could disconnect, reconnect, and resume without losing progress.
- Fan-out each run over **PubSub** so every client **observe()**s the same **runId** and the chat UI streams in realtime across all connected clients.
- Designed the message and tool-call UI so file context, workflows, and pause/resume stay in-thread instead of dumping as plain text.
- Built **workflows**, **memory systems**, schedules, tools, pause/resume, retries, sandboxing, and skills in **Mastra**; **Langfuse** for observability.
- **RAG**: **structure-aware** parsing and chunking, **metadata creation**, **retrieval funneling** with **Pinecone hybrid search** (dense and sparse vectors, BM25) and reranking; **SQL agent** for experimental data analysis.
- Enforced document-level permissions with **ABAC** based on document ownership and user roles and access levels.
- Ran on **Hono**, **Better Auth**, **Postgres** with **Drizzle** ORM; **S3** for file storage; deployed to AWS ECS Fargate using **SST** as IaC.
- Built a custom **@mentions-style** input component for attaching workspace files to messages (published on npm as **@ocwilsonchow/editor**).

Healthcare AI Agent

July 2025 – Apr 2026

Healthcare and shopping assistant grounded in uploaded documents and Shopify store product data.

- Designed the chat and admin UX in **Figma** and owned the client-facing flows — document upload, in-chat questionnaire, product and cart actions.
- Hardened Next.js admin and client apps with **Better Auth**, HttpOnly cookies, CORS, and RBAC.
- Streamed **LangGraph** agent replies into the chat with the **AI SDK** over tRPC, rendering tool calls as product cards, cart actions, and questionnaires instead of plain text.
- Built an admin form builder with conditional logic, modeled with **React Hook Form**, **Zod** and **XState**.
- Optimized expensive components with **memoization** and **useTransition** for heavy UI updates.

- Used **TanStack Query** with server prefetch and client-side rehydration for data lists, and **TanStack Table** with pagination, filters, and sorting; stored table state in URL search parameters so lists stay shareable and bookmarkable.

High-Performance Editorial Platform

2025

Led frontend development for a content and search editorial platform built with Strapi CMS, Meilisearch, and **Next.js**.

- Latest 100 articles statically rendered at build time, leaving the rest incrementally rendered on demand. Strapi **GraphQL** queries tagged properly with **Next.js** cache tags; revalidated through **webhook** triggered by content update events.
- Shipped instant search feature using Meilisearch with full-text search and fuzzy matching; result hits in **< 50ms**.
- PageSpeed Insights average scores **>90**; Reserved image space to prevent layout shifts and used **caching** and **image optimization** (enforced webp / avif formats and compression to **< 100 KB**) to reduce LCP: **FCP < 0.3s, LCP < 0.5s, CLS < 0.001**.
- Static assets cached at **CloudFront** to reduce latency and improve SEO.
- Improved **web accessibility** with keyboard-accessible navigation, accessible form-error handling, semantic structure, and Lighthouse / axe checks as release requirements rather than post-launch fixes.
- Built a **Strapi plugin** for AI content generation with **AI SDK**.

React Native Skia App

2024

- Owned the photobooth app from design and development through release on the Apple App Store and Google Play.
- Users take a selfie, load it onto the canvas, edit with stickers and pen tools, then upload and share.
- Built the canvas with **React Native Skia**, **React Native Reanimated**, and gesture handlers; captured images with **react-native-vision-camera**.
- Converted SVG stickers into Skia SVG React components, enabling **resize, rotate, pan, and scale** on the canvas.
- Managed state with **Zustand**; kept selections atomic and debounced high-frequency updates.
- Optimized with **memoization** and context isolation; held **55+ FPS** in RN Perf Monitor with over 100 stickers and strokes on the canvas.

React Native Student App

2023

- Owned student flows (course catalog, booking, announcements) from Figma through release on the Apple App Store and Google Play.
- Maintained **99%** crash-free sessions after launch by diagnosing crashes, device compatibility issues, and API bottlenecks with **Sentry**.
- Data fetched with **Tanstack Query**, cut duplicate fetches with cache and staleTime; client state managed with **Redux Toolkit**.
- Reduced feed scroll jank after profiling identified JS-thread stalls from long lists and full-resolution images; migrated lists to **FlashList** with row memoization, cached and downscaled feed images, and ran **Reanimated** transitions and gesture handlers on the native thread.
- Froze inactive tabs and blurred screens, and deferred mounts, to prevent unnecessary re-renders.
- Routed promo and milestone notifications to the matching screen via Firebase Cloud Messaging deep links.

OTHER WORK

Also shipped restaurant loyalty/student/membership apps (React Native), a homebound elderly care app (Flutter), campaign / e-commerce storefronts (Next.js, Shopify **GraphQL**), CMS content sites, and multiple RBAC admin systems.

SKILLS

Frontend: TypeScript, React, Next.js, React Native, Tailwind CSS, shadcn/ui, Figma, Zustand, Redux, TipTap

UI behavior: TanStack Query, TanStack Form, React Hook Form, XState, TanStack Table, next-intl

Architecture: state modeling (URL / server / local), design systems, tokens, React composition, component APIs, routing, SSR/ISR, GraphQL, tRPC, OpenAPI, Storybook, Fumadocs

Backend: Node.js, Hono, Drizzle, Prisma, OpenAPI, tRPC, Postgres, DynamoDB, Better Auth, Python FastAPI

AWS & DevOps: SST, Docker, GitHub Actions, AWS, Vercel, Cloudflare

Quality: Vitest, Playwright, Cypress, React Testing Library, Lighthouse, axe, Sentry

Mobile: Expo, FlashList, Reanimated, Moti, Skia, Unistyles, FCM

QUALIFICATIONS

- AWS Certified Solutions Architect – Associate, 2026
- AWS Certified Cloud Practitioner, 2025

EDUCATION

University College London

2016 – 2020

- Master of Pharmacy (MPharm, undergraduate) — First Class Honours
- Transitioned from pharmacy to software development after finding passion in building applications.